

and the eyewitnesses are *Crysis*, *Company Of Heroes* and *World In Conflict*.

There's another very important development worthy of mention. When thinking DX10, people usually think *costly*. Don't! The new generation of DX10 cards are much cheaper than you think. Although we've only looked at the high-end and mid-range cards this month, we're pleased to report that the new mid-range cards are actually the high-end parts of five months ago. Did you think NVIDIAs GeForce 9800GTX was the top-end successor to their older 8800GTX? You're right — it was. So what's its price, Rs 30,000? Wrong. This card is now priced at Rs 12,000, although it was priced at Rs 30,000 a scant four months back. Why the change? Competition. ATI has hit back hard, with its new Radeon HD4870 and HD4850, which have made quite a splash and have succeeded in winning back some much needed respect.

It seems parallelism is the in-thing, with NVIDIA upping its shader count to 240 units from 128 units, and ATI jumping to 800 units from a previous 320 units. GPUs are moving towards running more general purpose tasks like offloading physics and decoding from the CPU, thereby freeing up resources for other tasks. Nobody doubts the power of today's GPUs. With teraflop performance on a 55nm thick silicon wafer GPUs have come a long way from the checkerboard like multi-GPU graphics cards of yesteryear.

Before we get into the test we'd like to inform you that the entry level segment of GPUs will be covered in next month's issue. So watch out for it.



XFX GX-280-ZDD9

High-end Graphics Cards

For years the high-end graphics cards have held gaming and gamers in good stead. Nothing beats the experience of running a game at its highest quality settings, where you get all the eye candy unadulterated, and at blistering frame rates. NVIDIA and ATI have been slogging it out in this segment, although the former has been ruling the roost for the past two years. This was due to one GPU — the G80, better known as the 8800GTX, which spawned a host of other winners — the 8800GT, 8600GT and so on. Since November 2006, the 8800GTX has also dominated proceedings in all our graphics intensive tests, thanks to its monolithic architecture — an unheard of, (till then), 128 stream processors and a huge 384-bit wide memory bus.

The high-end segment in India is still niche. After all, who has Rs 30,000 or so to blow on a graphics card? A small and ever increasing number of gamers and the like are slowly adopting these high-end graphics cards into their PCs, but as a percentage such users are so insignificant

that most vendors here prefer to play the numbers game. One of the most noteworthy changes is the near equalisation of US and Indian prices — the gap is still large, but it continues to close. This benefits the high-end cards the most and we're pleased to report even the costliest of cards now costing under Rs 30,000. Up to even a year back, a high-end offering would command a price of around Rs 40,000 — preposterous really, especially considering the average Indian probably earns half of that. Let's look at the high-end offerings one by one.

NVIDIAs GeForce GTX 280

The sledgehammer strikes

The guys in green don't kid around with their GPU architectures. Neither have they been sparing the horsepower when it comes to creating über powerful GPU cores. The GTX 280 has a decidedly Bugatti Veyron approach to its processing power, and just like the car it goes all out — a near 90 per cent in SPs over its predecessor, the 8800GTX, a 512-bit memory bus connected to a whopping 1 GB of GDDR3 RAM, clocked at just above 2.2 GHz. All this makes the

High-end GPUs	NVIDIA GTX 280	NVIDIA GTX 260	NVIDIA 9800 GX2	ATI HD 4870 x2	ATI HD 4870	ATI HD 3870 x2
Stream Processors	240	192	128 x 2	800 x 2	800	320 x 2
Texture Addressing / Filtering	80 / 80	64 / 64	128 / 128	40	40	16 x 2
ROPs	32	28	32	16 x 2	16	16
Core Clock Speed (MHz)	602	576	600	750	750	775
Memory Bus Width (bits)	512	448	256 x 2	256 x 2	256	256 x 2
Memory Frequency (MHz)	2214	1998	2000	3600	3600	2250
Memory Type	GDDR3	GDDR3	GDDR3	GDDR5	GDDR5	GDDR4
Amount of memory (MB)	1024	896	512 x 2	2048	512	1024



Palit HD 4870 SONIC

GTX280 extremely capable and the brute force approach works well for NVIDIA — the GTX 280 is furiously fast, as our tests will show. In fact it's the fastest single GPU card on the planet.

Features

We received four cards based on the GTX 280 GPU — one each from ASUS, ZOTAC, XFX and POV. All these cards sport the default NVIDIA reference coolers with their own fancy, colourful stickers. Except for the POV GTX 280, the others were all factory over-clocked to 670 MHz core speeds, 602 MHz being the default clock for this GPU. Shockingly, only ZOTAC bundled a game with their card — GRID is a reasonably new game. They also bundle an HDMI adapter, as does POV. ASUS and XFX, the two old players miss out on a trick here.

Performance

At similar clocks, as expected, the three GTX 280s from ASUS, XFX and ZOTAC perform quite similarly, and there's a very small advantage going to the XFX card for some reason; but to be honest this advantage is never more than 1.5 per cent in all the games. The POV GTX 280 cannot keep up with this overclocked trio, and understandably trails them a small margin. Of particular interest is *Crysis*, which is taxing at anything above 1600 x 1200 pixels with DX10 effects turned on. None of these cards could reach the magic 30 fps

mark — shocking. If you want comfortable frames, you will have to lower your game resolution. *UT3* proves to be CPU limited with such powerful cards, and any of these are overkill for this game. *S.T.A.L.K.E.R.* proves no match for the 240 SPs working in tandem, and yields 100+ fps for all practical resolutions. *Company Of Heroes* is also very playable, and will give 100+ fps at any resolution. *World In Conflict* is another CPU limited game, and we recommend a quad core processor — our GTX 280s weren't able to crunch out any more than 35 fps. *F.E.A.R.* doesn't give the GTX 280 any issues, but this is an older game that scales quite well, and we were surprised seeing it slowing these cards to just above 100+ fps. For the record, XFXs GX-280-ZDD9 is the fastest GTX 280 card in the comparison.

NVIDIAs GeForce GTX 260 / GTX 260 216 core

Sledgehammer junior!

A notch lower is the GTX 260, which is a 192 SP part. It has 896 MB of GDDR3 RAM running at 2 GHz, and is fed by a 448-bit memory bus. Its core runs at a slightly lower 576 MHz (nearly identical as the old 8800GTX which ran at 575 MHz). With such on-paper specifications the GTX 260 is expected to be fast. A new addition to the GTX 260 family is the GTX 260 216 core edition. As the name suggests this part has 216 SPs instead of 192 SPs. Initially, we thought these were new 55nm parts, but they are based on the tried and tested 65nm fabrication process. Here's how NVIDIA gets 216 cores — a GTX 280 has its 240 SPs

grouped into ten distinct clusters of 24 SPs each. This cluster is called a TPC or Texture/Processor Cluster. Disable two clusters and you get 192 SPs — a GTX 260. Disable one, and voila, you get 216 SPs. The best silicon yields birth GTX 280s, while the lower binned silicon produces GTX 260s. Incidentally the GTX 260 and the GTX 260 216 core are clocked identically.

Features

We managed to get a ZOTAC GTX 260² in our hands, which is basically based on the new 216 SP design. The other two GTX 260s were 192 SP based models, one each from XFX and ZOTAC. The ZOTAC card was their AMP! Edition, which signifies aggressive overlocks. This card was clocked at 650 MHz — a significant hike over 576 MHz. The XFX card ran at stock speeds. All cards were on reference NVIDIA coolers. XFXs GX-260N-ADF9 ships with *Assassins Creed* which is a popular game, while ZOTACs GTX 260 ships with *GRID*. The ZOTAC GTX 260² doesn't ship with any game as of now, but we're told this card is spanking new and the in all probability a game will be bundled with retail versions of this card.

Performance

In our opinion the GTX 260² closes the gap between the GTX 260 and the GTX 280, and this card responds to overclocking quite well. It can come quite close to a stock clocked GTX 280 in some of the tests with the core at around 650 MHz. We expected the extra 24 SPs on the ZOTAC GTX 260² to push it close to their GTX 260 AMP! Edition. This doesn't happen at stock speeds. It seems the GTX 260 architecture benefits a lot from core speed increments, and the ZOTAC GTX 260 AMP! Edition was ahead by some margin. XFX's offering, the GX-260N-ADF9 trailed by a good bit, mainly because it was set at stock speeds. Frames in *Crysis* dip dangerously southwards of the magic 30 fps mark, even at 1280 x 1024 — this game is a real GPU